ONLINE COMPUTER STORE

Project on

"**Online Computer Store**"

In partial Fulfillment of the Requirement for Award of the Degree of

BACHLOR OF BUSINESS ADMINISTRATOR

(COMPUTER APPLICATION)

Mr. Mane Rohit Bapu **Roll No: - 39** **Exam Seat No: -**

Mr. Takale Partik Anil **Roll No: - 80** **Exam seat No: -**

Project Guide: - Mr. Anil Kalokhe

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ARTS SCIENCE & COMMERCE COLLEGE,

VIDYANAGARI, BARAMATI – 413133

SAVITRIBAI PHULE PUNE UNIVERSITY, PUNE

TYBBA(CA) (SEM-V) 2025-2026

VIDYA PRATISHTHAN'S

ARTS, SCIENCE & COMMERCE COLLEGE

VIDYANAGARI, BARAMATI -413133

CERTIFICATE

DEPARTMENT OF COMPUTER APPLICATION

This is to certify that **Mane Rohit Bapu** and **Takale Pratik Anil TYBBA(CA) (SEM-V)** has satisfactory carried out the project work according to the Syllabus prescribed by the University of Pune in **COMPUTER APPLICATION** and this project represents his Bonafide work in the year 2025 to 2026.

Roll No:39 **Exam Seat No:**

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_____ _____

Batch In charge **H.O.D.**

Date: / / 2025 **Date: / / 2025**

_____ _____

Internal Examiner **External Examiner**

Date: / / 2025 **Date: / / 2025**

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Chapter no.1

INTRODUCTION

1.1) Abstract:-

The Online Computer Store is a web-based e-commerce platform designed to provide customers with a seamless shopping experience for computer systems, accessories, and related electronic products. The system allows users to browse and search products, view detailed specifications, compare options, and place orders securely. Integrated features such as product filtering, cart management, user authentication, and online payment ensure a smooth and efficient purchasing process. On the backend, administrators can manage product listings, update inventory, process orders, and monitor sales performance. This platform leverages modern web technologies to deliver a responsive, scalable, and user-friendly solution that bridges the gap between customers and computer hardware suppliers.

1.2) Introduction to System: -

The Online Computer Store is an advanced e-commerce application designed to simplify the process of purchasing computers, laptops, and related accessories. With the growing demand for digital shopping platforms, this system provides customers with a convenient way to explore, select, and purchase products from the comfort of their homes.

The system is divided into two primary modules: **User Module** and **Admin Module**.

The **User Module** enables customers to register, log in, search products, filter by brand or category, view detailed product descriptions, add items to the shopping cart, and place secure online orders. It also supports order tracking and review submissions to enhance customer engagement.

The **Admin Module** provides store administrators with tools to manage inventory, add or update product details, handle customer orders, and monitor sales reports.

1.3) Company Profile: -

Online Computer Store is a web-based retail platform dedicated to providing customers with a wide range of computer systems, laptops, peripherals, and accessories. The company focuses on delivering a convenient, secure, and user-

friendly online shopping experience, ensuring that customers can access the latest technology products at competitive prices.

1.4) Scope of the System: -

The scope of the **Online Computer Store** defines the overall boundaries, objectives, and functionalities of the system. It covers the services provided to customers, as well as the administrative features available for store management.

System Scope:

- Provides a **web-based platform** accessible from desktops, laptops, and mobile devices.
- Ensures **data security** using authentication and authorization mechanisms.
- Integrates with **cloud services** (e.g., Cloud nary) for product image storage.
- Offers **scalability** to accommodate an increasing number of users and products.
- Maintains a responsive and user-friendly interface for smooth navigation.

1.5) Proposed System: -

The **Online Computer Store** is proposed as a complete e-commerce platform that overcomes the limitations of traditional shopping methods and existing manual systems. The system aims to provide a reliable, efficient, and user-friendly solution for both customers and administrators.

Chapter No.2

ANALYSIS

2.1 FACT FINDING TECHNIQUES: -

To analyze and design the **Online Computer Store**, various **fact-finding techniques** were used to gather information about user requirements, system functionalities, and existing problems with traditional/manual methods. The techniques used are as follows:

There are four types of fact-finding techniques: -

2.1.1 Interview: -

We conduct this technique because interview is very important data gathering techniques as in this the analyst directly contacts system and the potential user of the proposed system.

The interviews are of two types

1 Structured interview

2. Unstructured interview

i) Structured interview: -

These interviews are conduct because structured interviews are those where the interviewee is asked a standard set of questions in a particular order. All interviewees are asked the same set of questions.

- Do you prefer shopping for computers and accessories online or in physical stores? Why?
- What features would you expect in an online computer store (e.g., product comparison, reviews, filters)?
- How important is secure online payment for you when shopping online?
- Which device do you usually use for online shopping (desktop, laptop, or mobile)?
- Would you like to track your order after purchase?

ii) Unstructured Interview: -

This interview we conduct on undertaking in a question-answer format. This is of a much more flexible nature than the structured interview and can be very rightly used to gather general information about the system.

Here the respondents are free to answer in their own words. In this way their views are not restricted. So, the interviewer gets a bigger area to further explore the issues pertaining to a problem.

For ex.

- Can you describe your experience when buying a computer or accessories online?
- What problems have you faced while purchasing from physical stores or existing websites?
- What features do you feel are missing in current online computer stores?
- If you could design an online store yourself, what would you include in it?
- How do you decide which computer or accessory to buy (reviews, specifications, price, brand)?

2.1.2 QUESTIONNAIRE: -

To gather detailed information from potential customers and administrators regarding the requirements of the **Online Computer Store**, a questionnaire was designed. This questionnaire helped in identifying user preferences, challenges, and expectations from the system.

A. Open ended questionnaire:

Open-ended questionnaires are useful to capture **feelings, opinions, general experiences, and problems** faced by users and administrators.

1. Can you describe the process you go through when selecting computers or accessories to purchase online?

A) Researching latest technology trends

B) Reading online reviews and specifications

C) Asking for recommendations from friends/experts

D) Comparing different brands and prices

2. How do you think the computer shopping experience has changed in recent years, and what trends are you most excited about?

A) Rise of online shopping platforms

B) Availability of diverse brands and models

C) Growing demand for budget-friendly devices

D) Increased importance of fast delivery and service

3. What types of customer feedback have you found most valuable in shaping product offerings in an online computer store?

A) Performance and specifications feedback

B) Price sensitivity and affordability

C) Quality and durability of products

D) Overall shopping experience (website usability, delivery, etc.)

4. How do you usually interact with an online computer store, and what makes you trust the platform?

A) Detailed product descriptions and images

B) Verified customer reviews and ratings

C) Secure payment options

D) Efficient customer support and return policies

5. What challenges have you faced while shopping for computers or accessories online, and how do you usually deal with them?

A) Difficulty in comparing multiple products

B) Concerns about product quality or originality

C) High shipping costs or delays

D) Lack of proper customer support after purchase

B. Close-Ended Questionnaire –

Close-ended questionnaires control the frame of reference by presenting predefined options for respondents to select.

1. Do you feel the current process of purchasing computers and accessories online needs improvement?

A) Yes B) No

2. Would you like an online computer store to offer discount coupons or promotional offers?

A) Yes B) No

3. Do you think having eco-friendly and energy-efficient computer products is important?

A) Yes B) No

4. Would you prefer live chat or customer support assistance while shopping online?

A) Yes B) No

5. Should an online computer store provide multiple secure payment options (Credit/Debit, UPI, PayPal, COD)?

A) Yes B) No

2.1.3 OBSERVATION: -

Observation is one of the most effective fact-finding techniques, as it helps to directly study user behavior, existing processes, and challenges in the shopping experience. By carefully observing both customers and administrators, important requirements for the Online Computer Store system were identified

2.1.3.1 RECORD VIEW: -

Record review is an important fact-finding technique that involves examining existing documents, reports, and records to gather useful information for system analysis and design. For the **Online Computer Store**, various records were reviewed to understand the existing processes, data flow, and requirements.

1. Sales Records:

- Past invoices and receipts were studied to understand the structure of sales transactions.
- Information such as product name, quantity, price, discount, tax, and total bill amount helped in designing the order and billing module.

2. Product Catalogs:

- Existing product catalogs and brochures were reviewed to identify the required product attributes (e.g., brand, model, specifications, price, and warranty).
- Helped in deciding the database fields for product management.

4. Customer Records:

- Customer details from registration books and old records were reviewed.
- Helped in identifying the necessary fields for customer profiles (e.g., name, email, phone, address, past purchases).

5. Financial & Payment Records:

- Payment records such as cash receipts, credit/debit card slips, and online payment confirmations were studied.
- Highlighted the need for multiple secure payment options in the proposed system.

2.2 Feasibility Study: -

Feasibility study carried out whether there is complex problem. The purpose of feasibility study to investigate deeply the recommended system. Feasibility study is carried out to describe and evaluate the proposed system.

The study will justify whether the project is feasible or not and whether it is worthwhile or not?

There are three aspects of feasibility study.

1. Technical feasibility

2. Economic feasibility

3. Operational feasibility

2.2.1 Technical feasibility: -

Technical feasibility evaluates whether the proposed **Online Computer Store** can be developed using the available hardware, software, and technical resources. It ensures that the system is technically practical and can be implemented with the chosen technologies.

Hardware Requirements: -

Processor: - Intel ® Core 2duo.

RAM :- 8 GB.

HDD:-400 GB.

Software Requirements:-

Operating System: - ANY

Front End: React.js, Tailwind CSS.

Back End: Node.js Express.js.

Database: MongoDB

2.2.2 Economical Feasibility: -

In an **Economic Feasibility** study, the cost and benefits of the project are analyzed. This involves a detailed examination of the expenses required for system development, including hardware, software, design, development, and operational costs. Afterward, it is determined whether the project will be financially beneficial for the organization.

The **Online Computer Store** has been developed using open-source technologies like **MongoDB, Express.js, React.js, and Node.js (MERN stack)**, which reduces the overall cost of development. Since most of the required tools are free and open-source, the system can be implemented in a short period with minimum investment.

There are two types of economical feasibilities:

1. **One-time cost: -**

   Software - 50000 RS

2. **Recurring cost: -**

   Service and maintains – 6000rs

2.2.3 Operational feasibility: -

Operational feasibility focuses on ensuring that the **Online Computer Store** works smoothly in real-world conditions without errors or complications. It measures how effectively the proposed system solves existing problems and supports the business objectives. This feasibility also ensures that the system provides **security, reliability, and usability** for both customers and administrators.

Chapter no.3

System Design

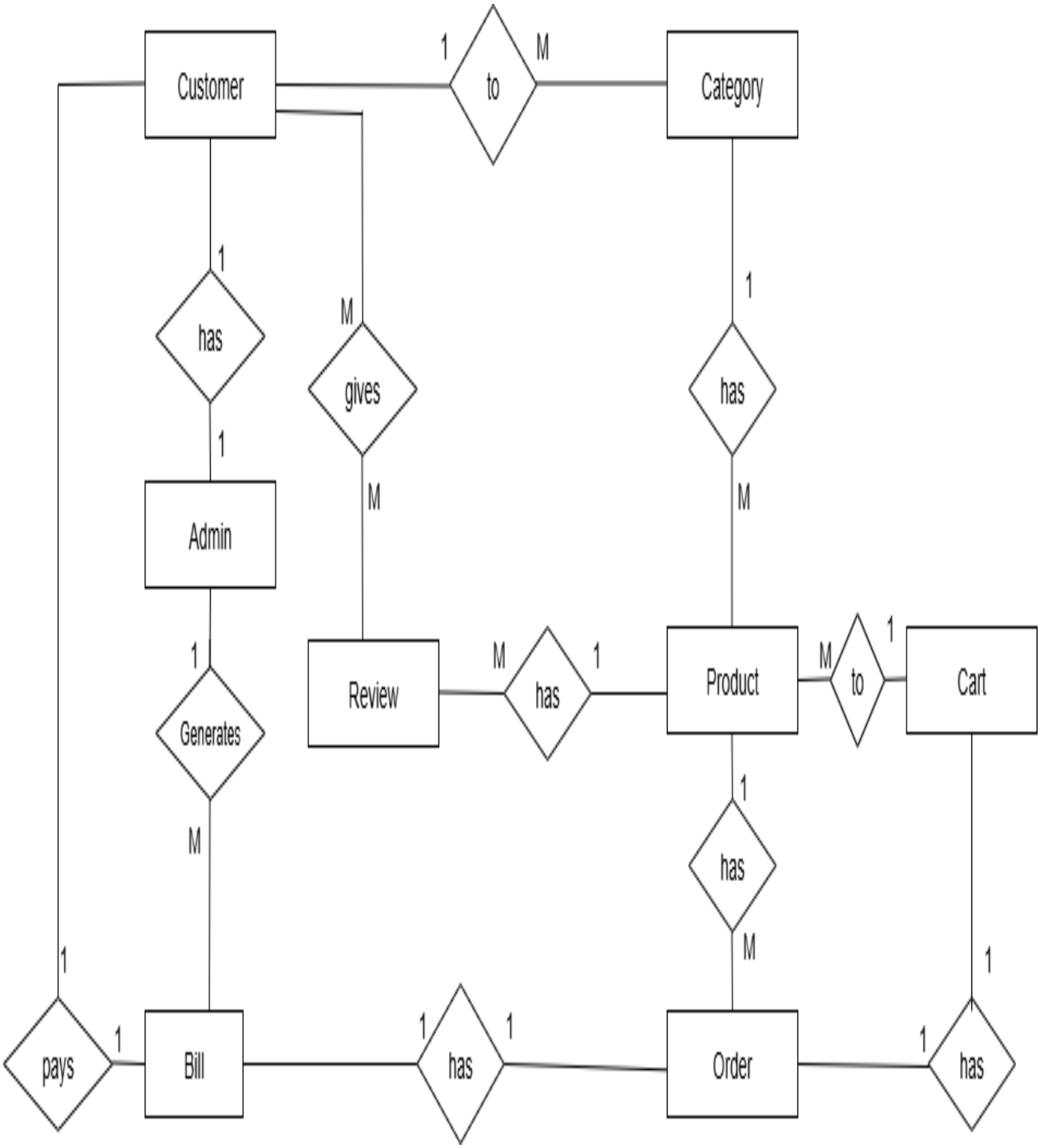
Entity & Attribute

Sr. no	Entity	Attribute
1	Customer	c_ id, c_ name, c_ address, c_ phone, c_ email
2	Admin	a_ id, a_ name, a_ email, a_ phone
3	Category	C_ id, cat_ name, cat _desc
4	Review	rid, rating, r_ text, r_ date, c_ id, p_ id
5	Product	p_ d, p_ name, p_ price, p_ model
6	Order	o_ id, o_ date, o_ status, o_ type
7	Bill	b_ no, b_ date, Gst _no, total
8	Cart	Cart_ id, qty, added _date, c_ status, c_ id, p_ id

3.2 Entity & Relationship

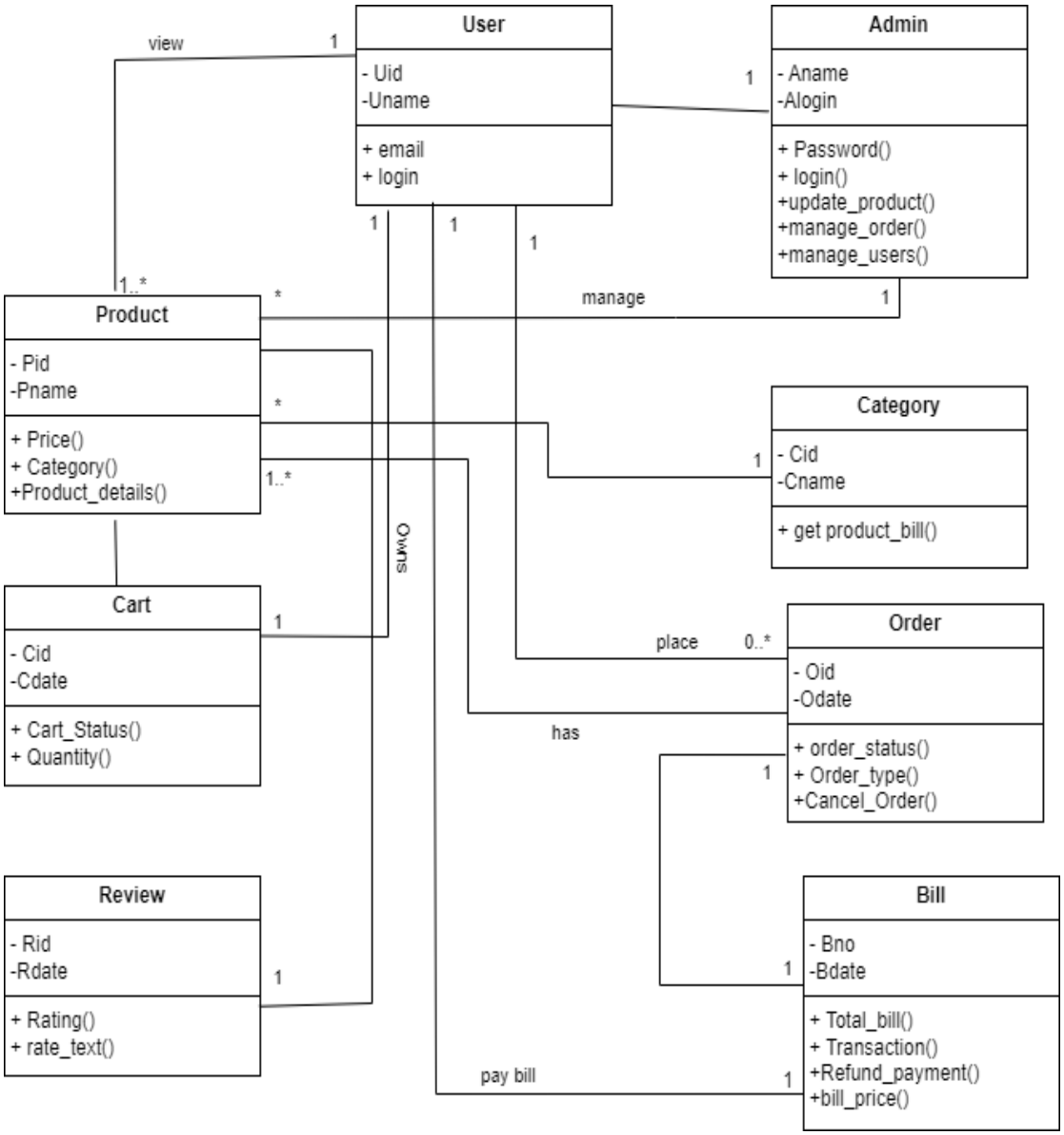
Sr. no	Entities	Relationship
1	Customer – Admin	One to One
2	Customer – Review	Many to Many
3	Customer – Category	One to Many
4	Customer – Bill	One to One
5	Admin – Bill	One to Many
6	Category – Product	One to Many
7	Product – Cart	Many to One
8	Product – Review	One to Many
9	Product – Order	One to Many
10	Cart – Order	One to One
11	Order – Bill	One to One

3.3 Entity Relationship

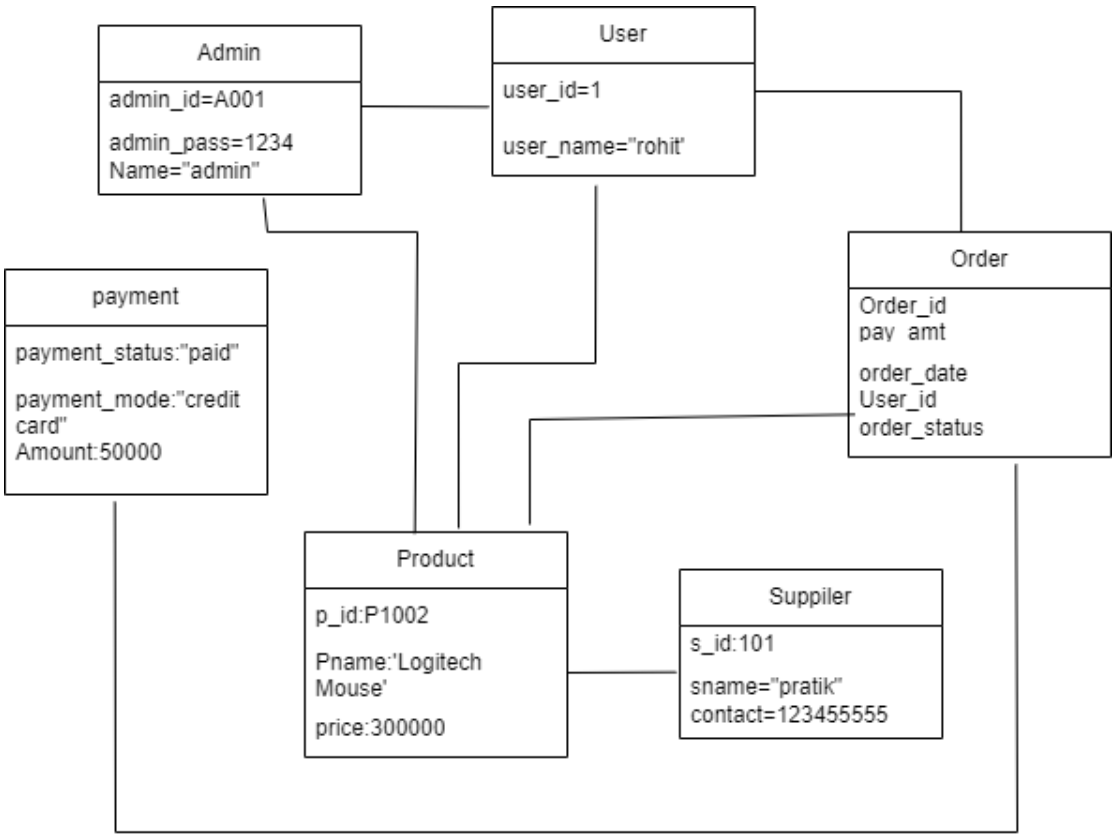


UML DIAGRAM

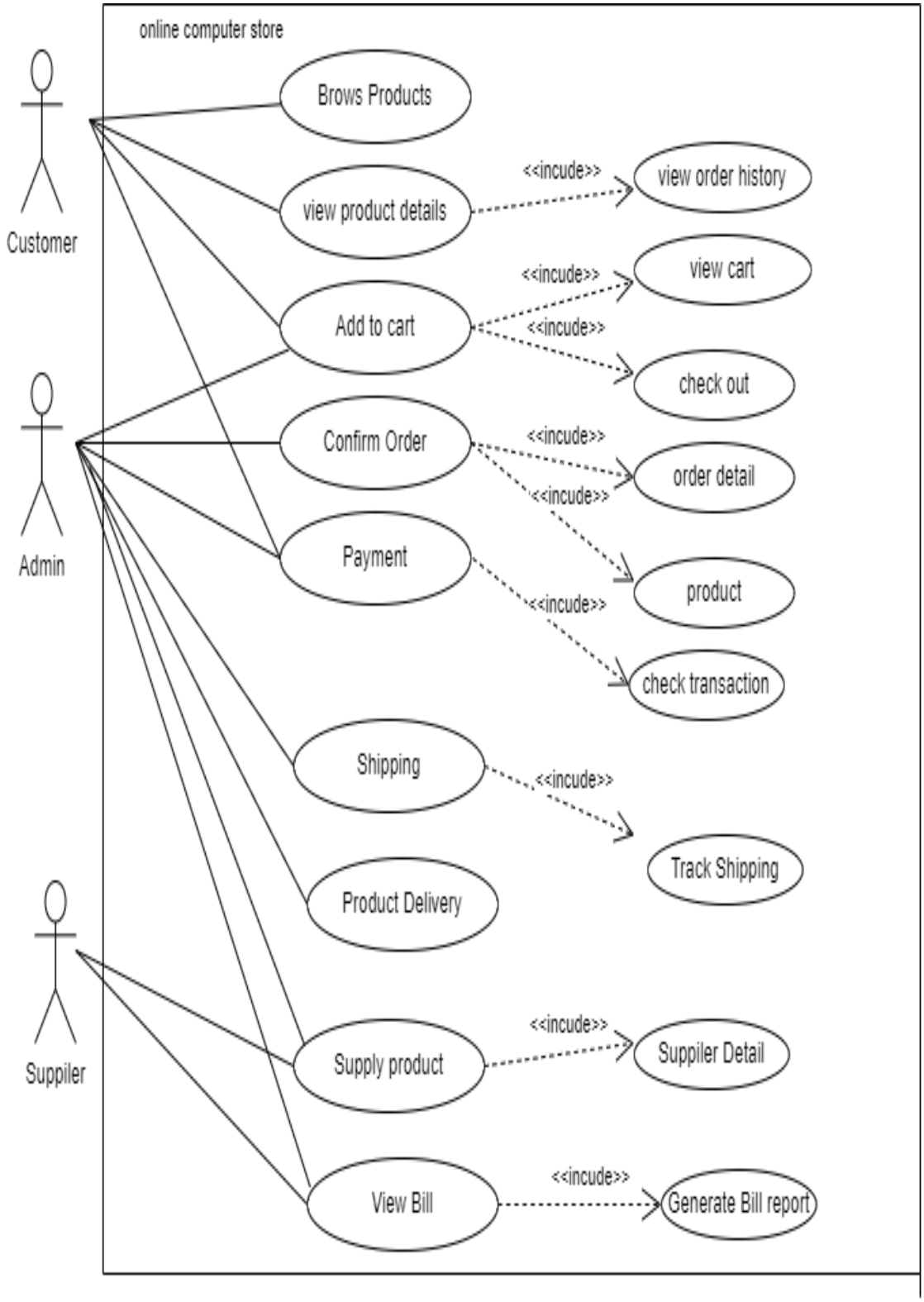
1.Class diagram:



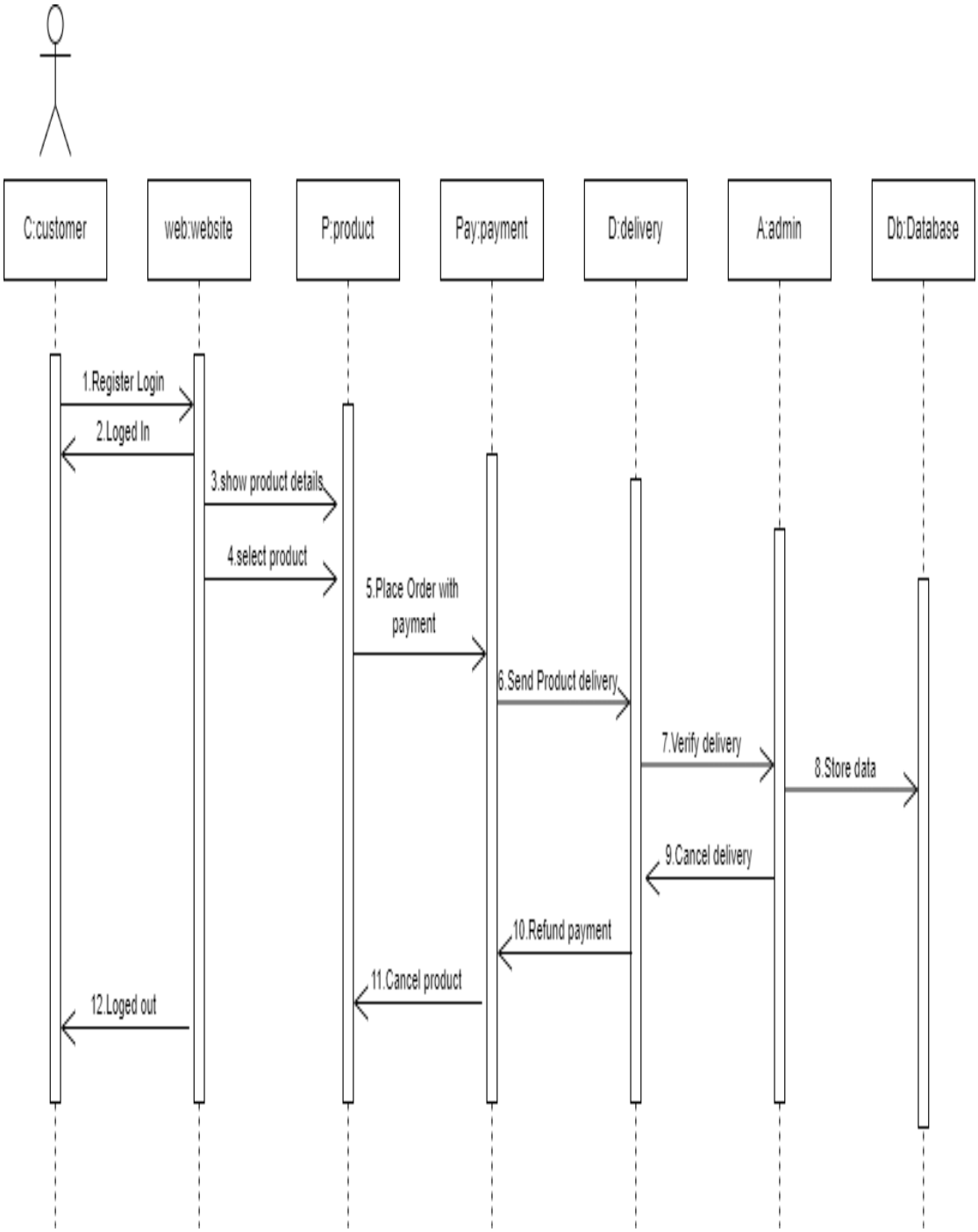
2.Object diagram:



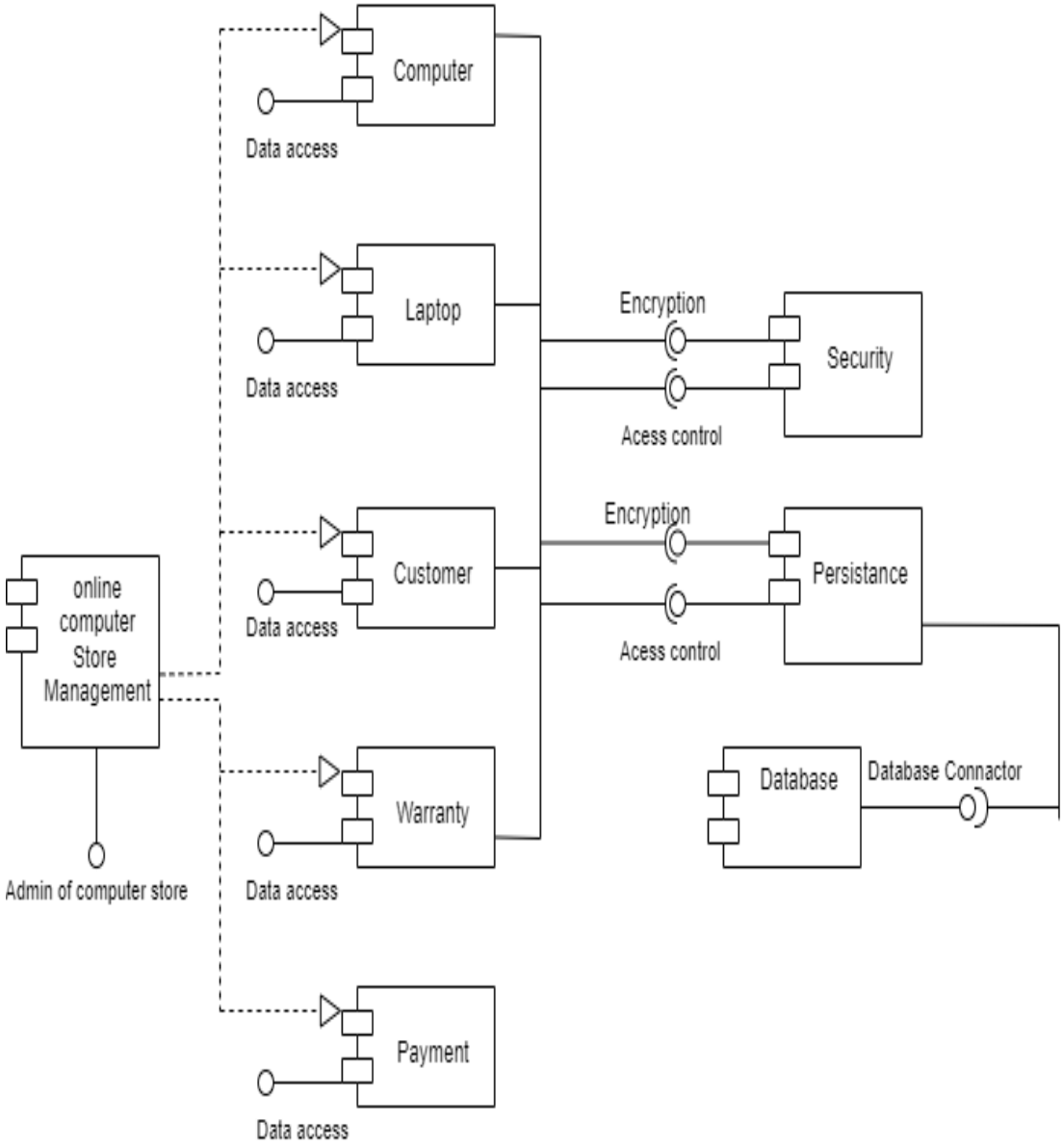
3.Use-Case Diagram



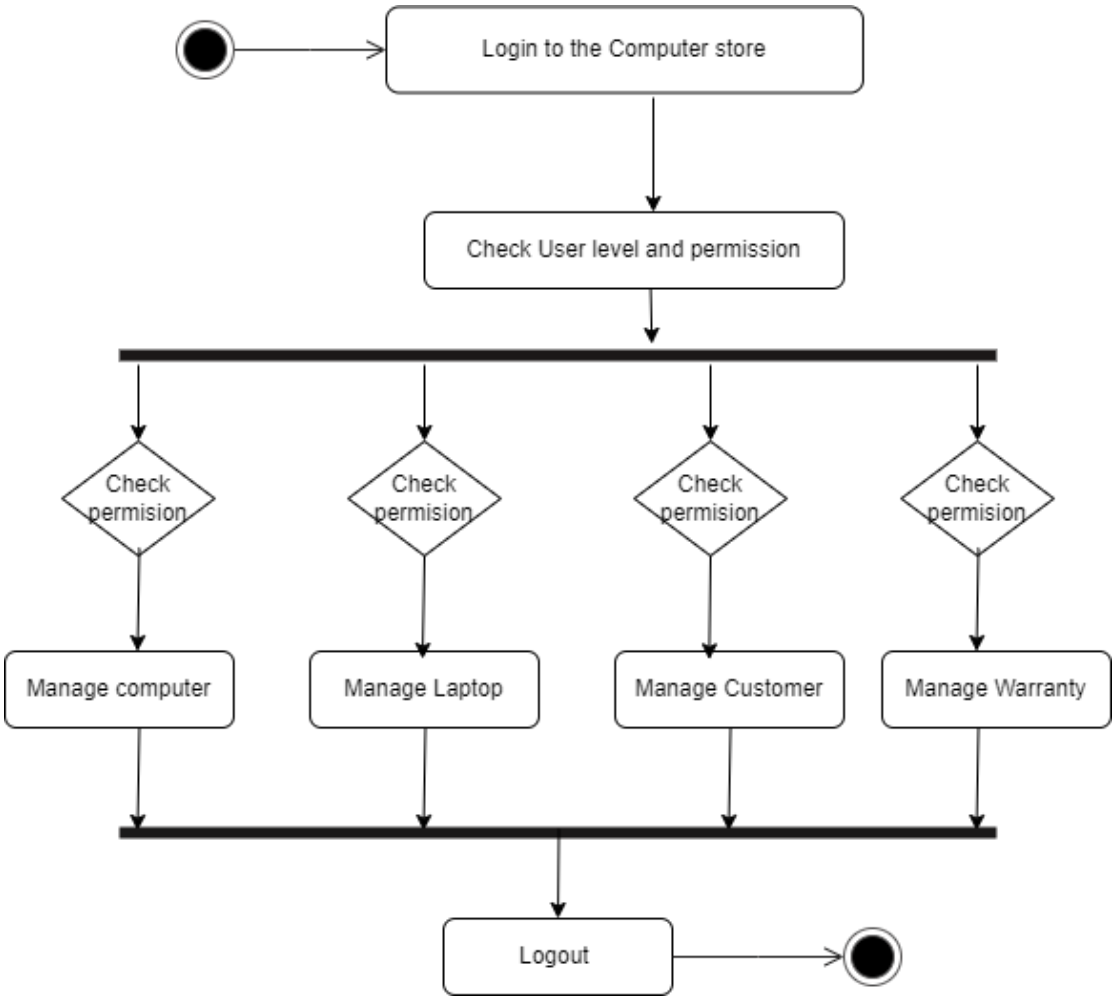
4.Sequence Diagram: -



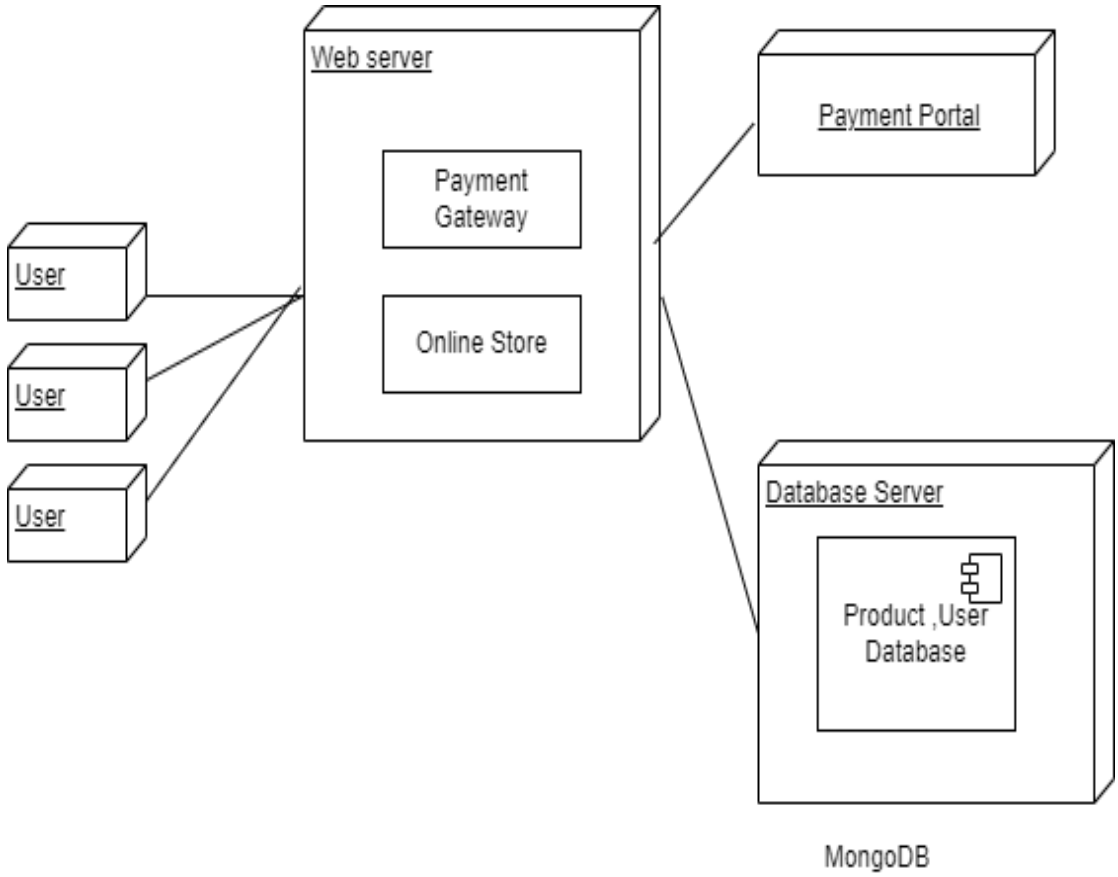
5.Component Diagram:-



6.Activity diagram: -



7.Deployment Diagram: -



Chapter no.4

File Design

Data Dictionary:

Field Name	Data Type	Size	Description	Constraints
C_ id	INT	5	Unique ID for each customer	Primary key, Auto Increment
C_ name	VARCHAR	15	Customer's full name	NOT NULL
C_ address	VARCHAR	20	Address of customer	NOT NULL
C_ phone	INT	10	Customer's phone number	NOT NULL, unique
A _id	INT	5	Unique ID for admin	Primary key, Auto Increment
A _name	VARCHAR	15	Name of admin	NOT NULL
A _email	VARCHAR	10	Email of admin	NOT NULL, unique
Cat_ id	INT	15	Unique category id	Primary key,Auto Increment
Cat_ name	VARCHAR	10	Category name/type	NOT NULL
Cat_ desc	VARCHAR	11	Category description	NOT NULL
rid	INT	15	Unique Review id	Primary key, Auto Increment

Field Name	Data Type	Size	Description	Constraints
rating	VARCHAR	5	Rating of product	NOT NULL
R_ text	VARCHAR	30	Review description	NOT NULL
R_ date	DATE	8	Review date	NOT NULL
P_ id	INT	3	Unique product ID	Primary key,Auto Increment
P_ name	VARCHAR	30	Product name	NOT NULL
P_ price	INT	5	Product price	NOT NULL
O_ id	INT	5	Unique order ID	Primary key, Auto Increment
O_ date	DATE	8	Date of order placement	NOT NULL
O_ status	VARCHAR	10	Order status	NOT NULL
B_ no	INT	5	Unique bill number	Primary key, Auto Increment
B_ date	DATE	8	Billing date	NOT NULL
Total	INT	5	Total bill amount	NOT NULL
Car_ tid	INT	5	Unique employee number	Primary key, Auto Increment
Qty	VARCHAR	20	Employee name	NOT NULL
Added_ date	INT	10	Employee mobile number	NOT NULL, unique

Field Name	Data Type	Size	Description	Constraint
C_ status	VARCHAR	20	Cart status	NOT NULL
C_ id	INT	5	Customer id	Foreign key constraint
P_ id	INT	3	Product id	Foreign key constraint

Data Flow :

Process Name	Description	Input Process	Output Process
Admin & Customer registration and Account Creation	System verifies credentials against the Login File	User sign-up	Account Created
Browse & Select Product	System manages & update stock and select product	Product Details & Manages Stock	Select Product & Updates Stock
Place order	System generates o_id, stores order details	Place An order	Billing generation
Bill generation	System calculates total amount and generates bill and report	Assist the bill report	Keep track of bill & report
Report Generation	This Process generates the Report File	Bill Details	Generate Report File

Data Store:

Data store name	Description	Data flow In	Data flow Out
Customer _muster	Stores permanent customer details	This flow has data flow in level 1.0	Data flow out in 5.0 for bill generation
Admin _muster	Stores admin user details	This file has data flow in level 2.0	Data flow out in 5.0 for bill generation
Account _muster	Stores account information	This flow has data flow in level 1.0	Data flow out in 5.0 for bill generation
Product_ muster	Stores details of all products	This flow has data flow in level 3.0	Data flow out in 5.0 for bill generation
Employee _muster	Stores employee details	This file has data flow in level 3.0	Data flow out in 5.0 for bill generation
Bill _muster	Stores bill details	This file has data flow in level 4.0	Data flow out in 5.0 for bill generation
Report _muster	Stores details of report	-	Data flow out in 5.0 for bill generation

Data Process:

Process Name	Description	Input Process	Output Process
Admin & Customer registration and Account Creation	System verifies credentials against the Login File	User sign-up	Account Created
Browse & Select Product	System manages & update stock and select product	Product Details & Manages Stock	Select Product & Updates Stock
Place order	System generates o_ id, stores order details	Place An order	Billing generation
Bill generation	System calculates total amount and generates bill and report	Assist the bill report	Keep track of bill & report

Chapter no.5

Table Design

Table Design: -

1]Customer:

Column Name	Data Type	Constraints
C_ id	INT	Primary Key, Auto Increment
C_ name	VARCHAR	NOT NULL
C_address	VARCHAR	NOT NULL
C_phone	VARCHAR	NOT NULL
C_email	VARCHAR	NOT NULL, Unique

2]Admin:

Column Name	Data Type	Constraints
A_ id	INT	Primary Key, Auto Increment
A_ name	VARCHAR	NOT NULL
A_ email	VARCHAR	NOT NULL, Unique
A_ phone	VARCHAR	NOT NULL

3]Category:

Column Name	Data Type	Constraints
Cat_ id	INT	Primary key, Auto Increment
Cat name	VARCHAR	NOT NULL
Cat desc	VARCHAR	NOT NULL

4]Review:

Column Name	Data Type	Constraints
Rid	INT	Primary Key, Auto Increment
Rating	VARCHAR	NOT NULL
R_ text	VARCHAR	NOT NULL
R date	DATE	NOT NULL
C_ id	INT	Foreign Key constraint
P_ id	INT	Foreign Key constraint

5]Product:

Column Name	Data Type	Constraints
P_id	INT	Primary Key, Auto Increment
P_ name	VARCHAR	NOT NULL
P_ price	INT	NOT NULL
P_ model	INT	NOT NULL

6]Order:

Column Name	Data Type	Constraints
O _id	INT	Primary Key, Auto Increment
O _date	DATE	NOT NULL
O _status	VARCHAR	NOT NULL
O_ type	VARCHAR	NOT NULL

7]Bill:

Column Name	Data Type	Constraints
B_ no	INT	Primary Key, Auto Increment
B_ date	DATE	NOT NULL
Gst _no	INT	NOT NULL
Total	INT	NOT NULL

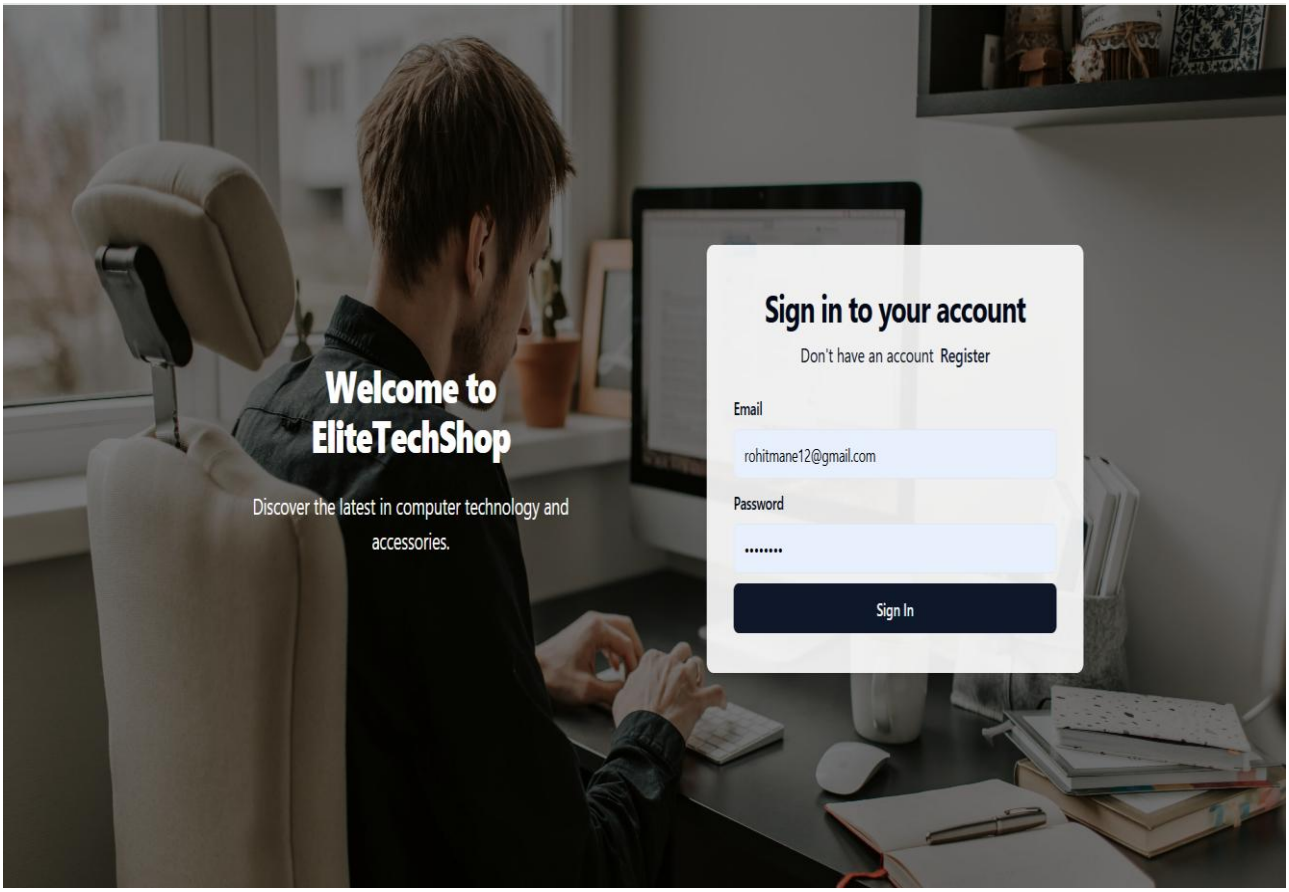
8]Cart:

Column Name	Data Type	Constraints
Cart_ id	INT	Primary Key, Auto Increment
qty	INT	NOT NULL
Added _date	DATE	NOT NULL
C_ status	VARCHAR	NOT NULL
C_ id	INT	Foreign Key constraint
P_ id	INT	Foreign Key constraint

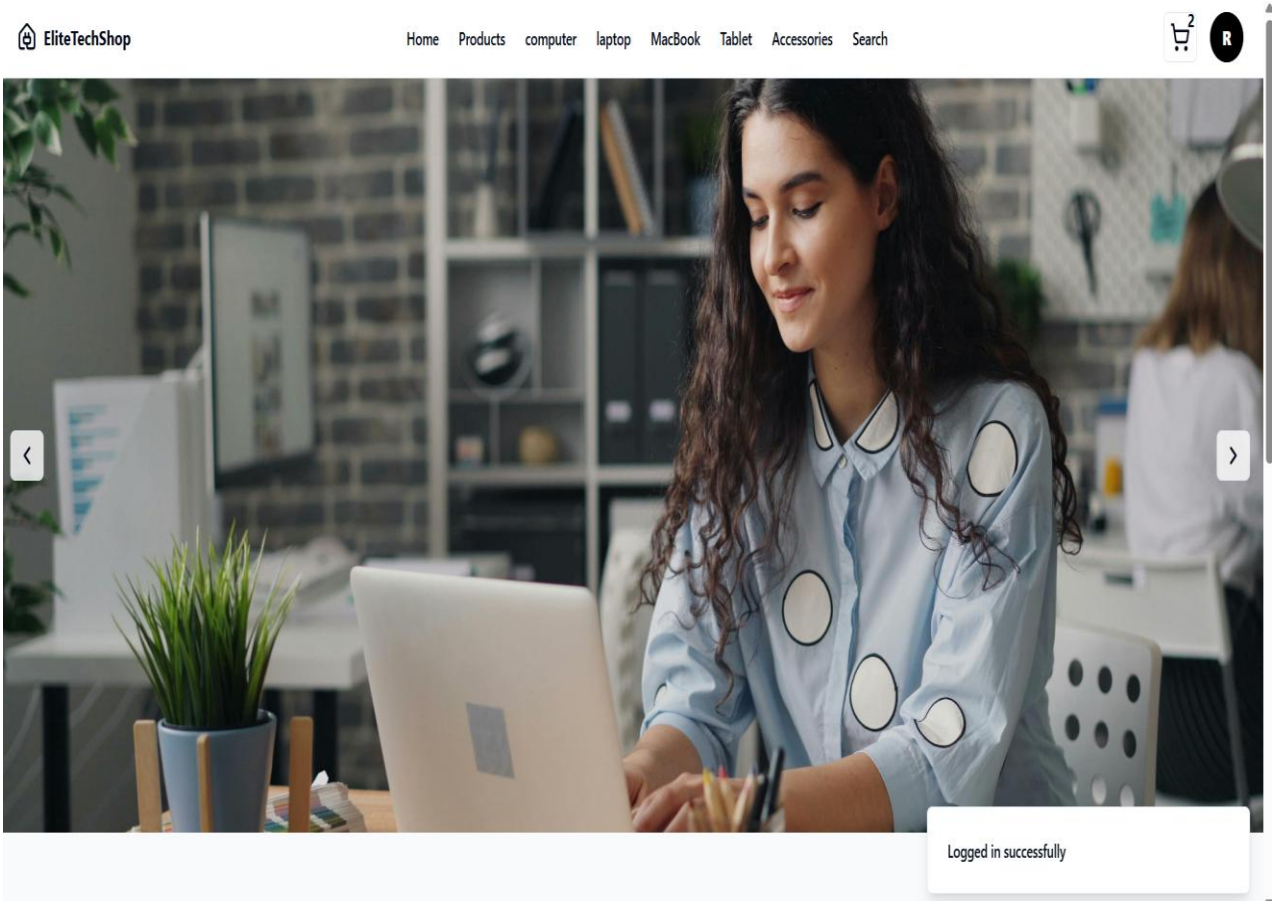
Chapter: - 6

Form Design and Report

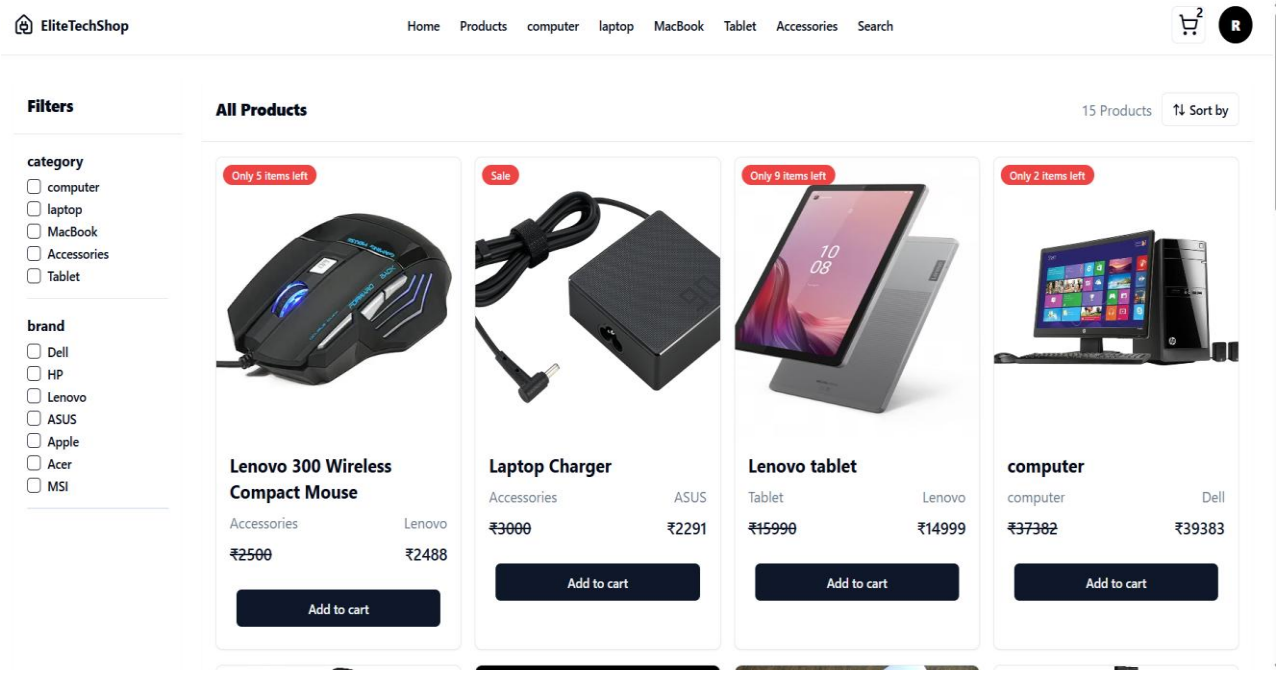
Sign page:-


Welcome to
EliteTechShop
Discover the latest in computer technology and accessories.
Sign in to your account
Don't have an account Register
Email
rohitmane12@gmail.com
Password
Sign In

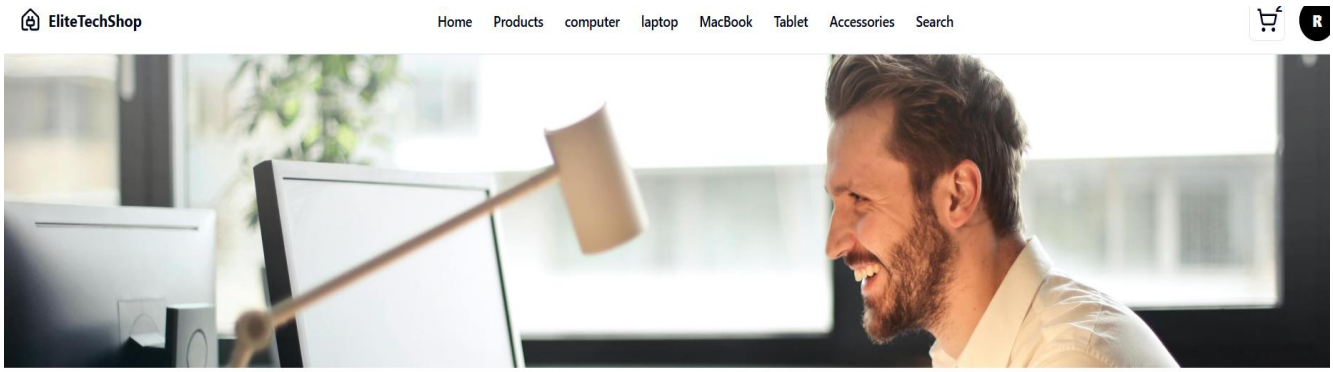
Home page :-


EliteTechShop
Home
Products
computer
laptop
MacBook
Tablet
Accessories
Search
2
R
Logged in successfully

Product page :-



User Account page:-

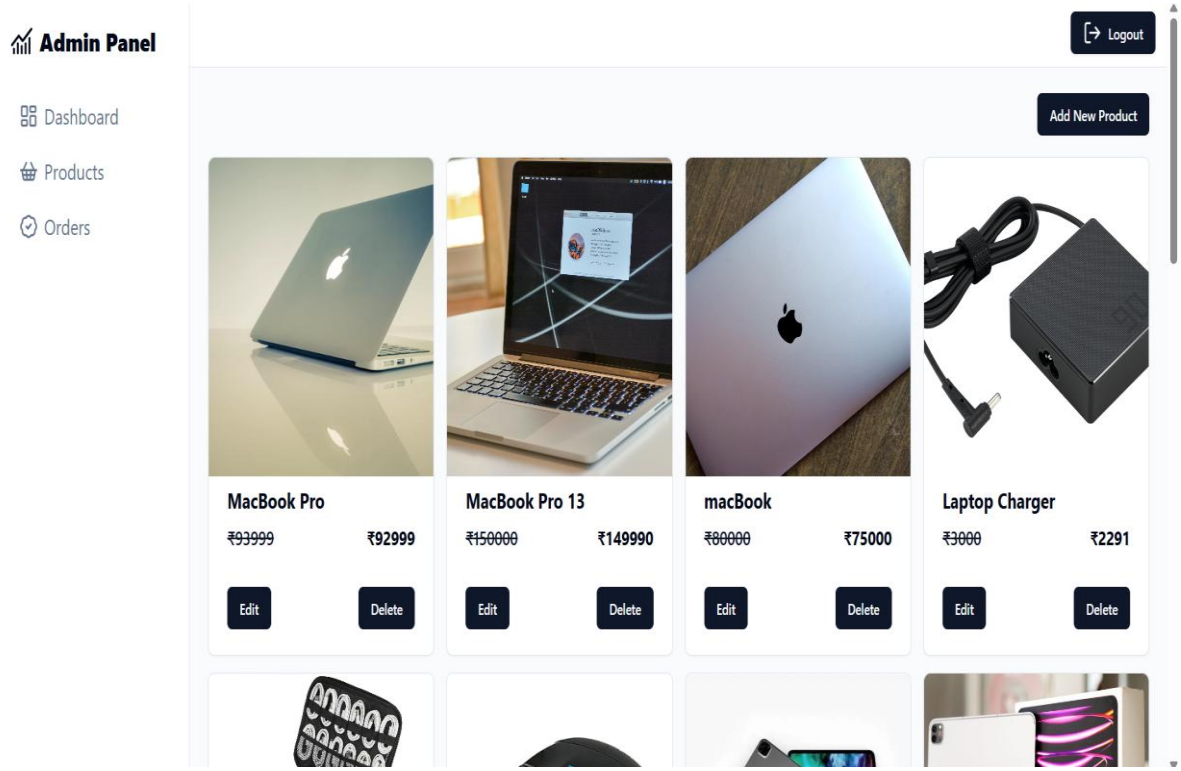


Orders Address

Order History

Order ID	Order Date	Order Status	Order Price	
68ca64dccb8668ba2b8df7ab	2025-09-17	inProcess	$766	View Details
68ca64ddcb8668ba2b8df7ae	2025-09-17	delivered	$766	View Details

Admin Panel page :-



Chapter no.7

Advantages & Disadvantages

Advantages –

- **Wide Product Range**
  Handles multiple categories like laptops, desktops, accessories, and computer components with detailed specifications.
- **Inventory Management**
  Automatically updates stock after each purchase, helping prevent out-of-stock or overstock situations.
- **Secure Online Payments**
  Integration with **PayPal and other gateways** ensures safe and reliable transactions for customers.
- **Customer Data Management**
  Stores customer profiles, order history, and preferences for personalized recommendations, offers, and loyalty programs.
- **Reports & Analytics**
  Provides detailed reports on sales, revenue, top-selling products, and customer trends to support business decisions.
- **Time-Saving & Convenience**
  Customers can shop 24/7 from anywhere without visiting a physical store, reducing effort and increasing satisfaction.

Disadvantages –

- **Initial Setup Cost**
  Developing and deploying a MERN-based online store with hosting, domain, and database setup can be costly initially.
- **Training Required**
  Admins and staff may require training to manage the backend, update products, and handle customer queries.
- **Maintenance & Updates**
  Regular technical maintenance, bug fixing, and feature updates require ongoing resources.

- **Dependence on Internet & Servers**
  The system depends on stable internet and hosting servers. Any downtime can disrupt sales operations.
- **Data Security Risks**
  Customer data, payment details, and business information must be secured against potential cyber threats and hacking attempts.
- **Technical Complexity**
  Compared to a physical store, running an online store requires technical expertise for deployment, scaling, and integration with third-party services.

Chapter no.8

Future Enhancement

Future Enhancement:

1. Advanced Inventory Management

2. Customer Relationship Management (CRM)

3. Online Store Integration

4. Multiple Payment Options

5. AI & Analytics

6. Multi-User & Role-Based Access

7. Mobile App Support

8. Cloud Backup & Security

9. Integration with Marketing Tools

10. Multi-Store Management

Chapter no.9

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